Read two numbers and perform arithmetic operation using switch:

```
Enter  two numbers   5   4
                    M    E    N     U
                    +. Add
                    -. Sub
                    *. Mul
                    %. Mod
                    /. Div
                    E. Exit
                    Enter ur option[ * ]
                                        Product = 20
```

```
#include<stdio.h>

#include<conio.h>

#include<math.h>

void main()

{
```

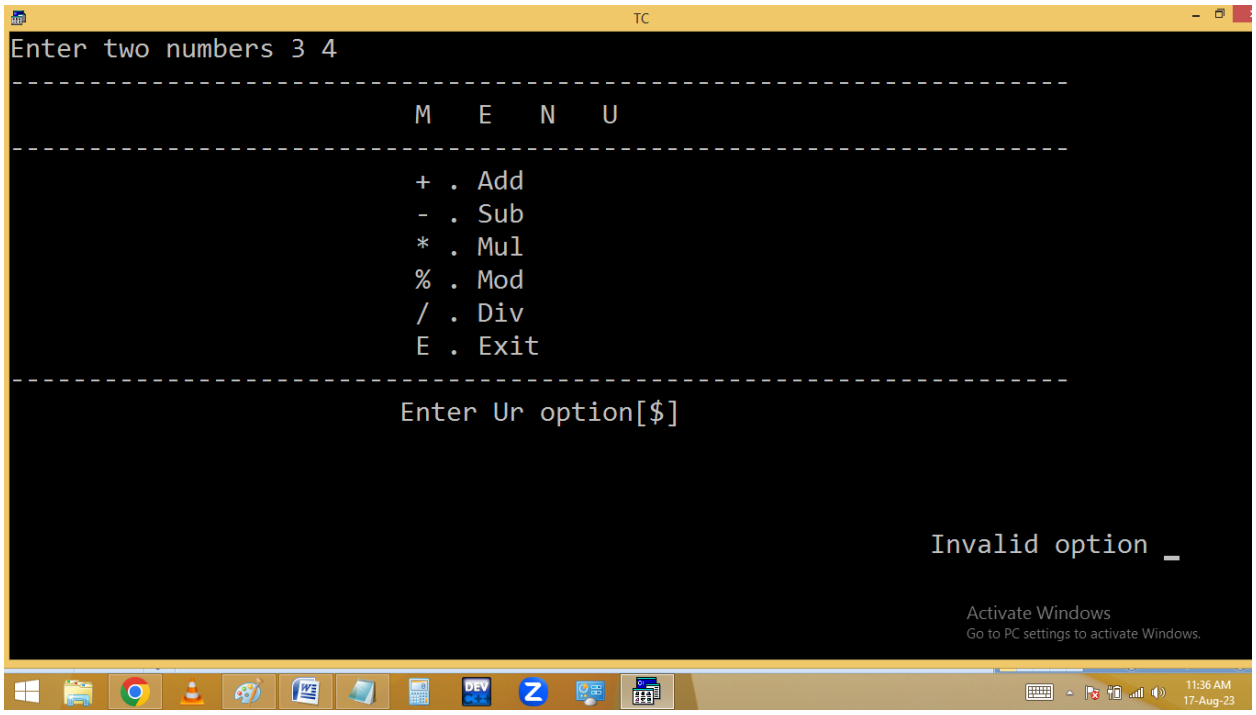
```
float a,b;
char  op;
abc:
clrscr();
printf("Enter two numbers "); scanf("%f %f",&a,
&b);
puts("-----------------------------------------------------------
-------");
puts("\t\t\t  M   E   N   U");
puts("-----------------------------------------------------------
-------");
puts("\t\t\t  + . Add");
puts("\t\t\t  - . Sub");
puts("\t\t\t  * . Mul");
puts("\t\t\t  % . Mod");
puts("\t\t\t  / . Div");
```

```
puts("\t\t\t  E . Exit");
puts("-----------------------------------------------------------
-------");
flushall();
printf("\t\t\t Enter Ur option[ ]\b\b");
scanf("%c",&op);
gotoxy(60,16);  /* col, row */
switch(op)
{
case  '+': printf("Sum=%.2f",a+b);break;
case  '-': printf("Sub=%.2f",a-b);break;
case  '*': printf("Mul=%.2f",a*b);break;
case  '%': printf("Mod=%.2f",fmod(a,b));break;
case  '/': printf("Div=%.2f",a/b);break;
case  'e': case 'E': return;
default:  printf("Invalid option ");
```

```
}
getch();
goto abc;
}
```

```
Enter two numbers 3.3  1.22
-------------------------------------------------------------------
                          M   E   N   U
-------------------------------------------------------------------
                         + . Add
                         - . Sub
                         * . Mul
                         % . Mod
                         / . Div
                         E . Exit
-------------------------------------------------------------------
                        Enter Ur option[/]



                                                        Div=2.70
```

```
Enter two numbers 6 7
-------------------------------------------------------------------
                          M   E   N   U
-------------------------------------------------------------------
                         + . Add
                         - . Sub
                         * . Mul
                         % . Mod
                         / . Div
                         E . Exit
-------------------------------------------------------------------
                        Enter Ur option[e]
```



Loops / Iterations / Repetitive statements

Loops are used to repeat a block/group of statements continuously until the given condition becomes false.

Loops reduce program size and improves performance.

In loops beginning and ending points are same.

C-Language supports basically 2 types of loops.

1. **Entry/pre controlled loops.**
2. **Exit/post controlled loops.**

In entry control loops, condition is tested first and it is true then only statements block is executed.

Under entry control loops we are having

i. **While loop**
ii. **For loop**

In exit control loop, the statements are executed first and later condition is tested.

Under exit control loop we are having

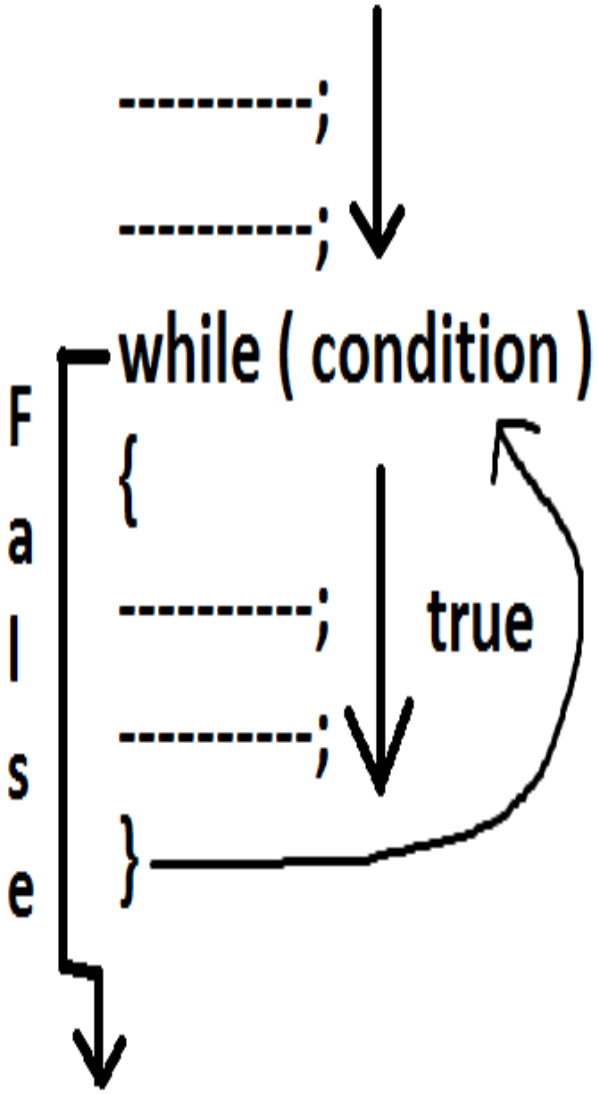
i. **do while.**

<u>While loop:</u>

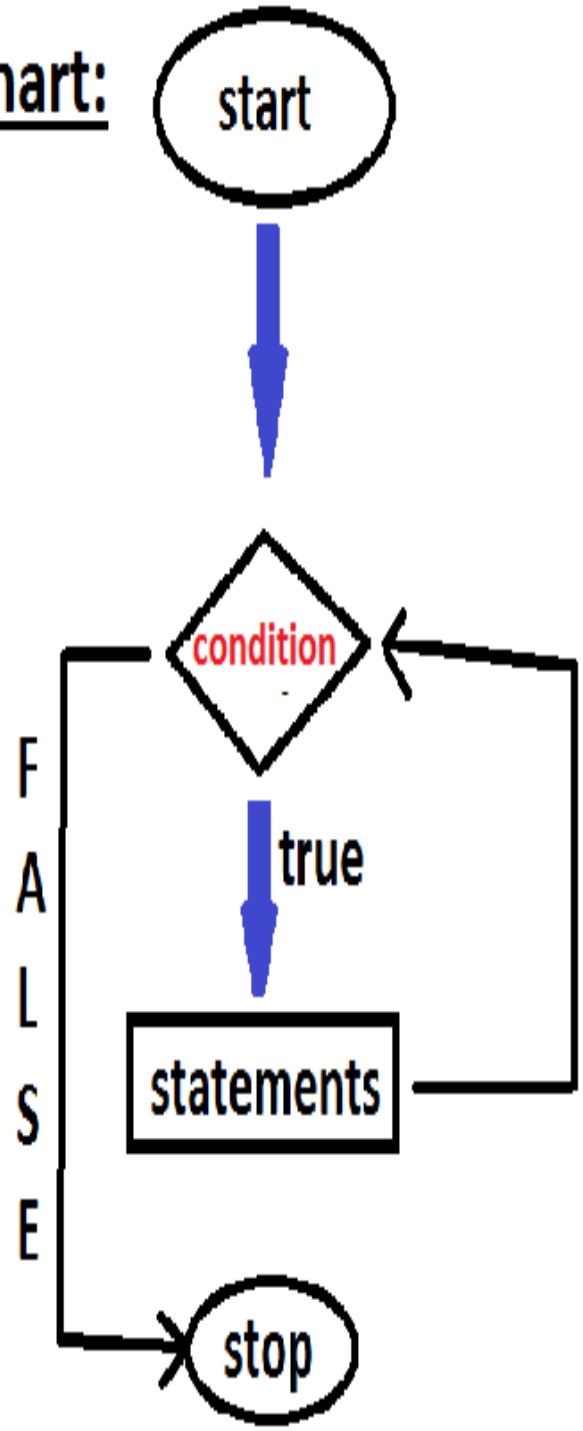
- **while is a keyword.**
- **In while loop condition is tested first and it is true then only while block statements are executed. After executing while block statements, the program execution automatically shifted/jumped to while condition at the beginning. If it is true then once again the while block statements are repeated. Like this the process is continued until while condition becomes false.**
- **While is entry control loop.**

Syntax:

```
----------;
----------;
while ( condition )
{
----------;
----------;
}
```



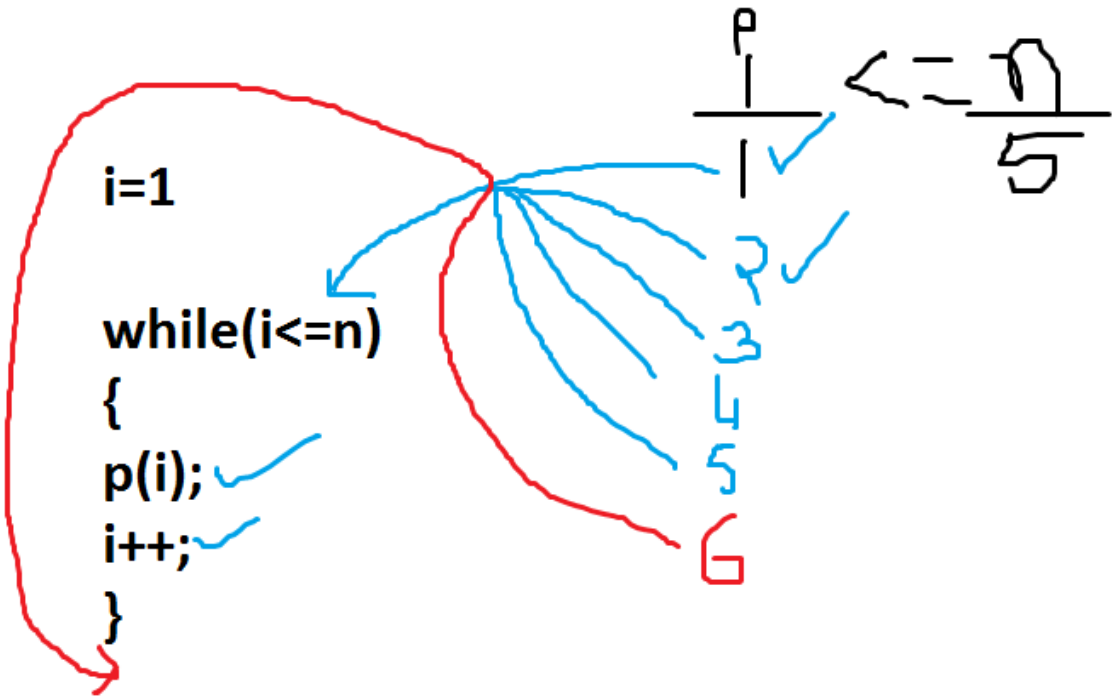
Flow chart:



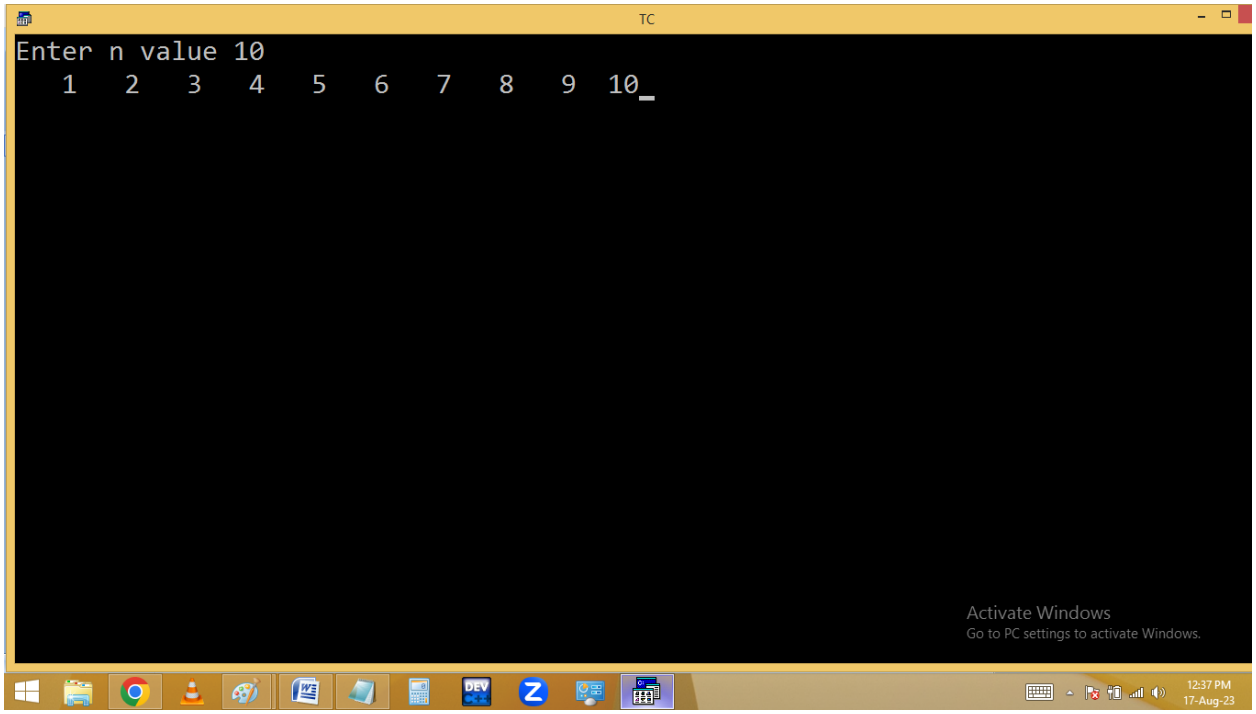
Printing 1..n natural numbers:

```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
     Line 10    Col 1   Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1, n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(i<=n)
{
printf("%4d",i);
i++;
}
getch();
}
```

```
Enter n value 100
   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  2
  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  4
  41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  6
  61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  8
  81  82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99 10
```




```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
      Line 10    Col 1    Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1, n;
clrscr();
printf("Enter n value "); scanf("%d",&n);

while(i<=n)printf("%4d",i++);

getch();
}
```


TC
Enter n value 10
1 2 3 4 5 6 7 8 9 10
Activate Windows
Go to PC settings to activate Windows.
12:37 PM
17-Aug-23

TC

File Edit Run Compile Project Options Debug Break/watch

Line 9 Col 22 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1, n;
clrscr();
printf("Enter n value "); scanf("%d",&n);

while(i<=n)printf("%-4d",i++);

getch();
}
```

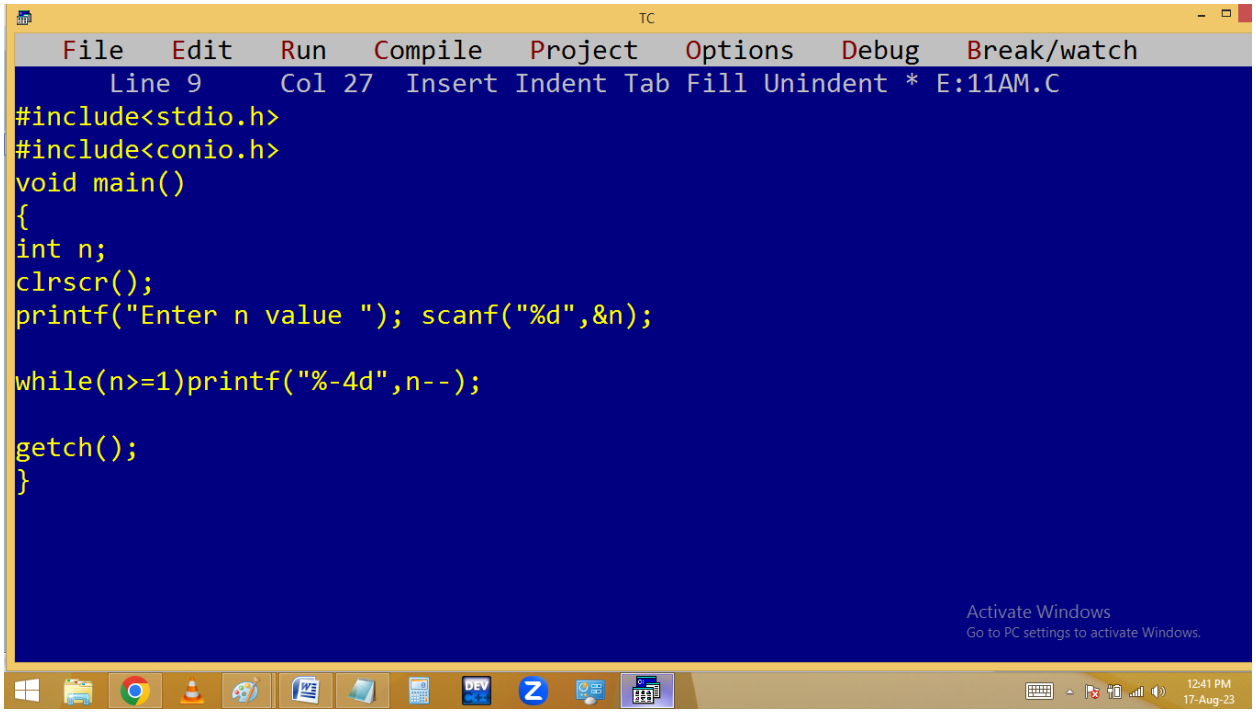
Activate Windows
Go to PC settings to activate Windows.

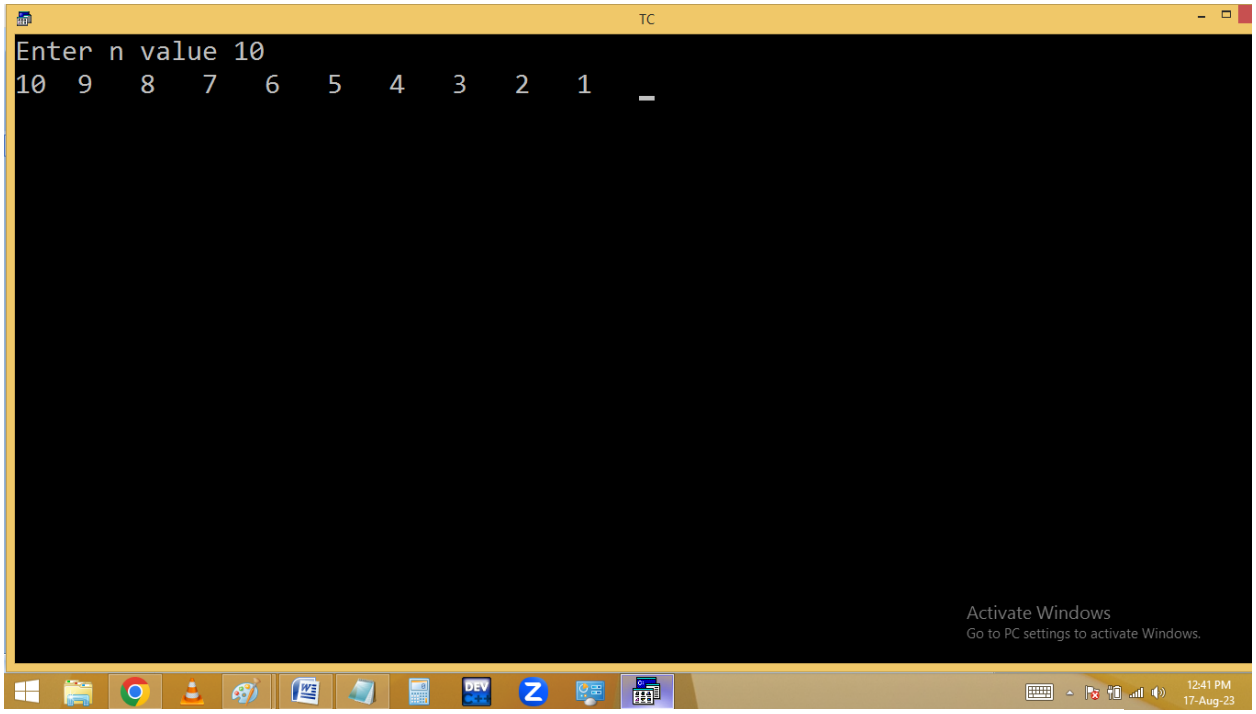
TC

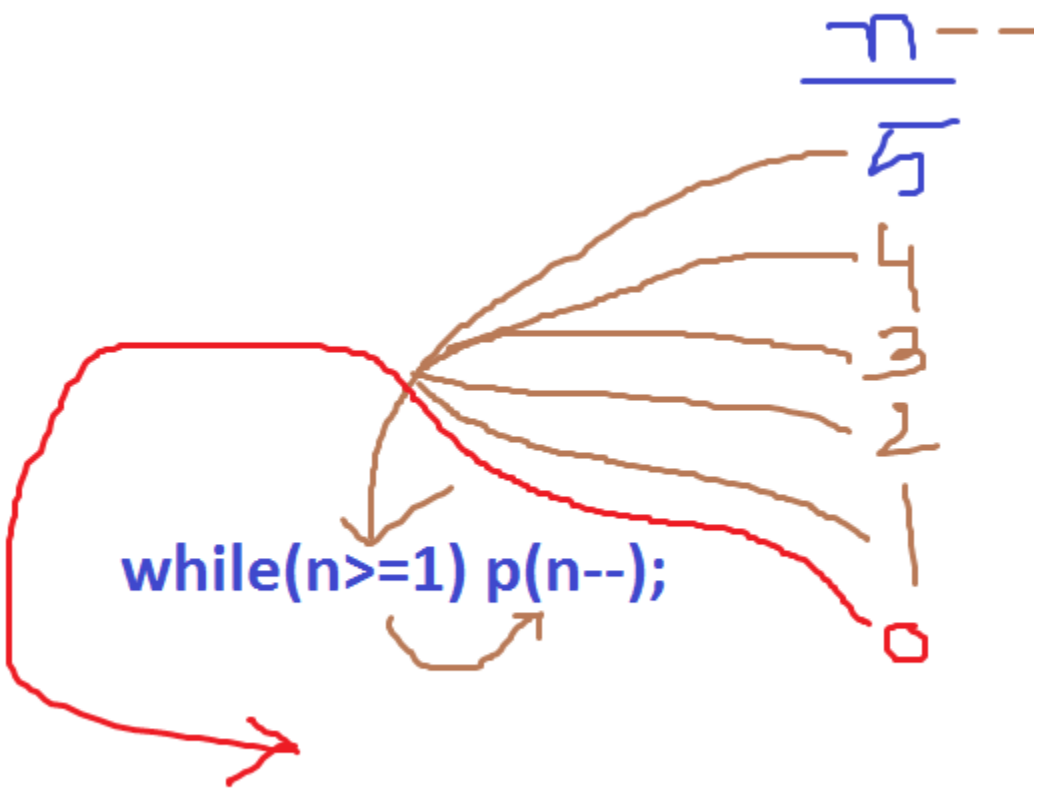
```
Enter n value 100
1   2   3   4   5   6   7   8   9   10  11  12  13  14  15  16  17  18  19  20
21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40
41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60
61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80
81  82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99  100
```

Activate Windows
Go to PC settings to activate Windows.

Printing 1..n numbers in reverse order.




TC
Enter n value 10
10 9 8 7 6 5 4 3 2 1
Activate Windows
Go to PC settings to activate Windows.
12:41 PM
17-Aug-23

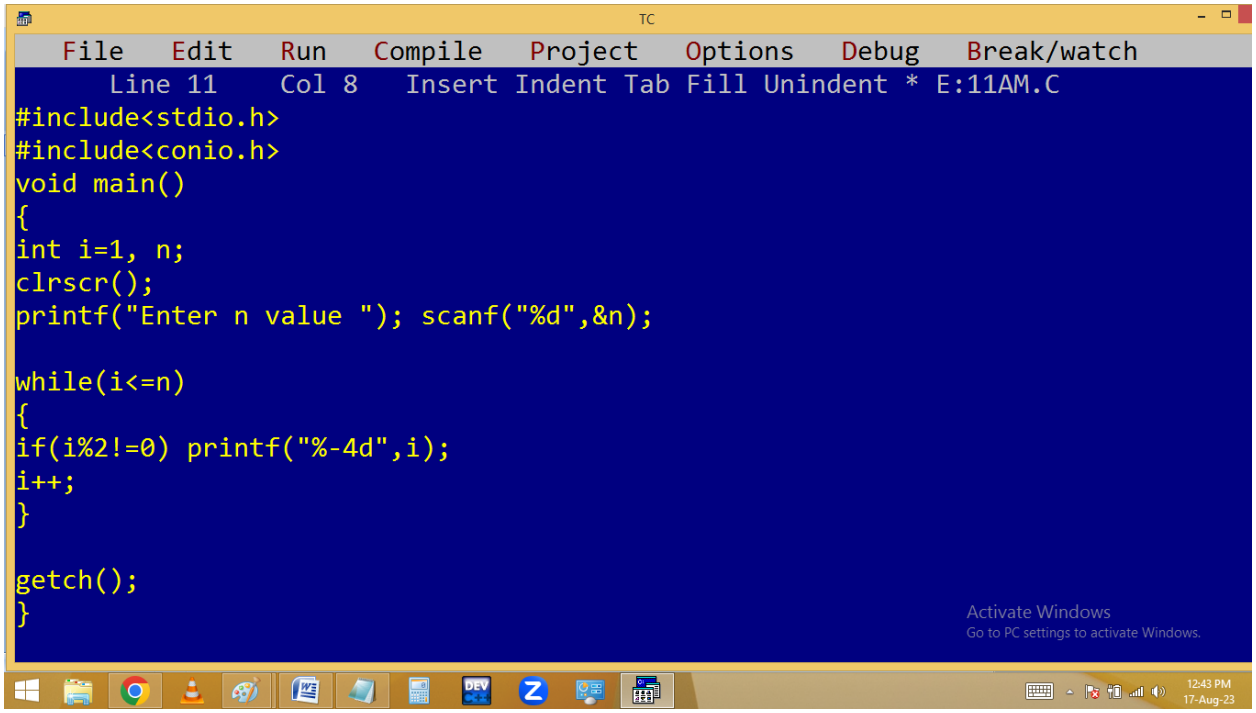

n--
5
4
3
2
1
0
while(n>=1) p(n--);

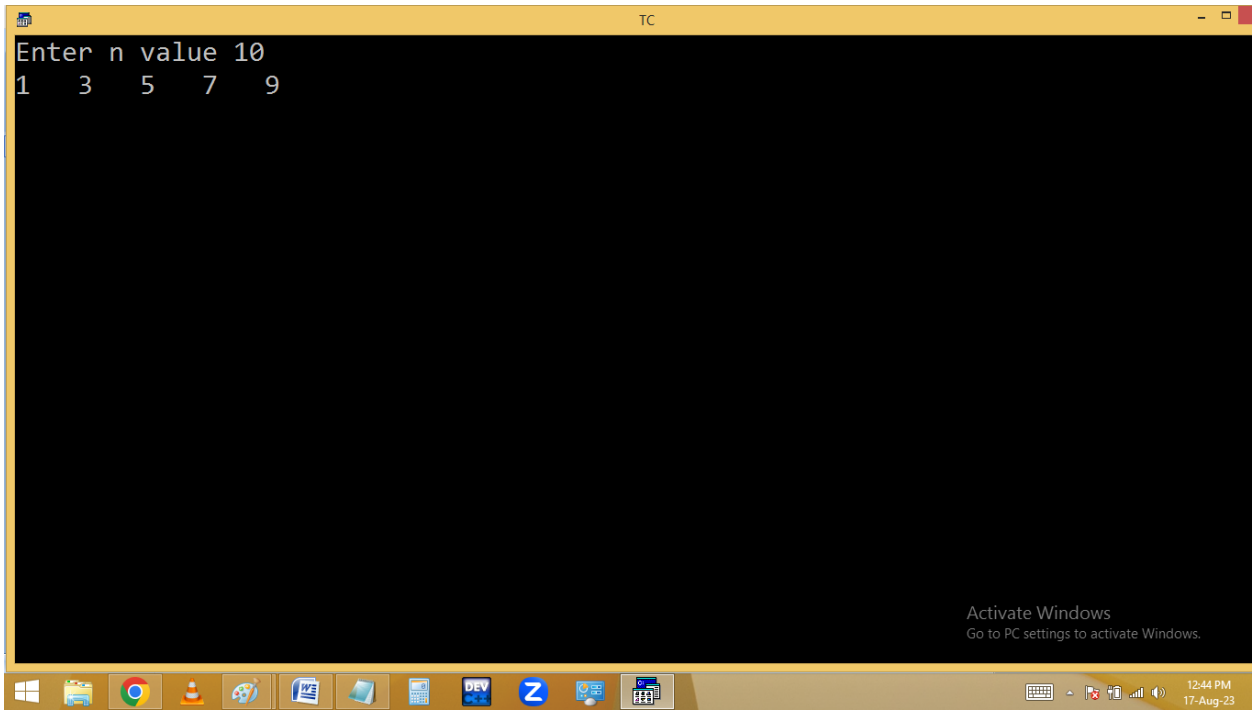
Printing 1..n even numbers.

```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
     Line 13    Col 2   Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1, n;
clrscr();
printf("Enter n value "); scanf("%d",&n);

while(i<=n)
{
if(i%2==0) printf("%-4d",i);
i++;
}

getch();
}
```

```
Enter n value 100
2   4   6   8   10  12  14  16  18  20  22  24  26  28  30  32  34  36  38  40
42  44  46  48  50  52  54  56  58  60  62  64  66  68  70  72  74  76  78  80
82  84  86  88  90  92  94  96  98  100
```

Eg. printing 1..n odd numbers





Home work:

Finding 1..n numbers sum and avg [mean].

1 to 5 number sum = 15

15/5=3 is avg

Find 1..n even, odd numbers sum

5 < 1+3+5=9 ✓

2+4 = 6 ✓